

BACAU, Romania

WMO: 151500

Lat: 46.5578N

Lon: 26.8967E

Elev: 575

StdP: 14.39

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.4	10.0	0.0	5.6	5.7	5.6	7.5	11.3	23.4	24.3	20.2	27.9	3.1	340	0.475

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.8	91.1	70.3	87.8	69.3	84.9	68.3	73.3	85.6	71.6	84.0	70.1	81.6	4.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.4	110.5	78.9	67.8	104.2	76.7	66.2	98.5	75.2	37.2	85.4	35.7	84.0	34.4	81.5	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.4	15.7	13.5	DB	-2.9	95.9	5.3	4.1	-6.7	98.8	-9.8	101.2	-12.8	103.5	-16.7	106.5	(4)
(5)			WB	-3.3	76.4	5.3	2.1	-7.1	77.9	-10.2	79.1	-13.2	80.3	-17.1	81.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.7	27.7	31.8	40.5	51.6	61.5	68.0	71.3	70.4	61.5	50.7	41.1	30.6		
	DBStd	17.07	9.18	9.88	8.62	6.83	6.27	5.31	4.91	5.23	6.41	7.48	8.06	9.12		
	HDD50	2544	691	509	309	61	4	0	0	0	2	84	283	601		
	HDD65	5808	1156	928	760	405	142	30	7	13	139	447	716	1065		
	CDD50	2785	0	1	14	108	360	540	661	633	347	104	17	0		
	CDD65	576	0	0	0	2	34	120	203	181	34	2	0	0		
	CDH74	6055	0	0	1	52	454	1238	2029	1840	407	34	0	0		
	CDH80	2126	0	0	0	11	96	396	790	730	100	3	0	0		
(14) Wind	WSAvg	4.9	5.4	5.8	6.5	5.9	4.9	4.2	3.9	3.7	4.0	4.0	5.2	4.9		
(15) Precipitation	PrecAvg	21.5	0.9	0.9	1.1	2.0	2.8	3.0	3.1	2.3	1.9	1.1	1.2	0.9		
	PrecMax	33.5	2.7	2.4	3.9	5.0	7.1	6.1	5.7	6.0	6.4	4.8	3.7	2.8		
	PrecMin	13.3	0.2	0.3	0.0	0.2	0.9	0.7	1.3	0.2	0.0	0.0	0.1	0.1		
	PrecStd	4.6	0.7	0.6	0.8	1.1	1.7	1.4	1.3	1.5	1.7	1.0	0.9	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.5	59.7	69.4	80.0	86.9	91.4	96.7	96.1	87.4	78.3	67.2	54.3		
		MCWB	44.8	48.6	52.3	60.9	64.7	71.2	70.1	70.5	66.0	62.7	55.6	46.6		
	2%	DB	46.9	53.5	63.6	73.5	82.5	87.5	91.4	91.0	82.6	72.4	60.6	48.8		
		MCWB	41.1	45.7	50.5	57.8	64.3	69.6	70.5	70.1	64.7	59.9	52.8	44.4		
	5%	DB	42.3	48.2	58.7	69.2	79.1	84.5	87.7	87.5	78.7	67.3	55.8	44.8		
		MCWB	38.9	42.8	48.5	55.4	63.1	68.0	70.1	69.4	63.7	57.6	50.9	41.1		
	10%	DB	39.0	44.3	54.4	65.0	75.5	81.5	84.3	84.0	74.7	62.8	52.2	41.3		
		MCWB	36.9	40.5	45.9	53.3	61.3	66.8	69.2	68.2	61.9	55.4	48.9	38.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.1	50.2	54.7	62.6	68.4	74.3	75.4	75.9	69.7	64.9	57.0	48.8		
		MCDB	51.6	57.9	65.2	78.4	81.3	87.0	87.5	87.9	82.3	75.3	64.3	51.8		
	2%	WB	41.9	46.3	51.7	59.4	66.3	71.8	73.3	73.1	66.9	61.5	54.3	44.9		
		MCDB	45.6	51.7	62.0	71.4	78.8	84.1	85.8	85.0	77.9	69.5	59.2	48.5		
	5%	WB	39.4	43.4	49.2	56.9	64.3	70.1	71.8	71.1	65.0	58.7	51.9	41.8		
		MCDB	42.0	47.3	57.7	66.6	76.1	81.5	84.0	83.5	75.8	65.9	55.3	44.1		
	10%	WB	36.9	40.8	46.7	54.5	62.4	68.1	70.2	69.4	63.0	56.2	49.3	39.1		
		MCDB	38.8	44.2	53.3	62.9	73.3	78.4	81.6	80.7	72.5	61.9	52.1	40.9		
(35) Mean Daily Temperature Range	MDBR		11.7	13.5	17.5	20.3	22.2	21.8	22.8	23.5	21.4	18.7	12.9	11.5		
	5% DB	MCDBR	17.4	20.6	26.4	28.6	28.1	27.8	28.9	30.0	28.9	26.8	20.9	17.0		
		MCWBR	12.3	13.7	15.3	14.5	12.8	12.2	11.5	11.8	12.9	14.9	13.4	12.4		
	5% WB	MCDBR	15.3	18.4	24.7	25.1	25.4	24.9	25.1	26.2	24.9	24.7	17.5	15.2		
		MCWBR	11.2	12.8	15.0	13.5	12.6	12.1	11.6	11.8	12.1	14.3	11.6	11.7		
(40) Clear-Sky Solar Irradiance	taub		0.323	0.353	0.392	0.453	0.426	0.424	0.448	0.458	0.413	0.376	0.348	0.321		
	taud		2.313	2.249	2.205	2.065	2.220	2.264	2.206	2.169	2.255	2.338	2.362	2.339		
	Ebn at Noon		239	254	261	254	267	268	259	251	251	243	228	225		
	Edh at Noon		24	32	39	49	43	42	44	44	37	29	23	22		
(44) All-Sky Solar Radiation	RadAvg		405	649	1024	1388	1781	1868	1892	1718	1223	786	421	323		
	RadStd		52	81	103	126	155	168	158	129	126	92	38	34		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.90	N/A	N/A	+1.46	N/A	N/A	N/A	N/A	+130	+81
(47) Regional (49 neighbors)		+1.12	N/A	N/A	+1.28	N/A	N/A	-248	-329	+130	+29

Nomenclature: See separate page